OPH Cosmology Data and Likelihood Contracts: Frozen Sources, No-Data-Use Receipts, and Comparable Predictions

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Abstract

OPH cosmology can be compared with the sky only after its inputs are fixed. This paper states the comparison contract for CMB, BAO, supernova, weak-lensing, growth, and large-scale-structure data.

A close visual overlay or a good binned curve is a diagnostic. A prediction requires source data fixed before the comparison, declared solver settings, a frozen likelihood run, pooled statistics, and a falsification rule. The point is to keep a simulator resemblance from being promoted into a sky prediction by name alone.

1 Scope

This is a technical companion outside the release bundle. It sets the data and likelihood boundary for CMB/CAMB comparisons, other cosmological data products, and visualizer payload provenance emitted by the OPH-FPE simulator.

2 Shared Finite Quotient and Claim Tiers

Finite quotient ensemble theorem surface. Fix a finite regulator r. The physical presentation space is Σ_r, the presentation redundancy groupoid is Γ_r, and the finite physical quotient is

$$Q_r = \Sigma_r/\Gamma_r, \qquad \pi_r : \Sigma_r \to Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map n_r is a normal-form map, not a probability law. Any promoted physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

Observable algebras and reference states. In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra $\mathcal{A}_r^{\mathrm{phys}}$ with state

$$\omega_r(A) = \mathrm{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$\left(\mathcal{A}_{r,b}^{\mathrm{phys}}, \tau_{r,b}^0\right), \qquad \mathcal{A}_{r,b}^{\mathrm{phys}} = z_{r,b} B(\widetilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where $U_r : G_r \to U(\widetilde{\mathcal{H}}_r)$ is the compact gauge action and $z_{r,b}$ is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(A)}{\mathrm{Tr}_{\widetilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b}\widetilde{\mathcal{H}}_r \cong \bigoplus_\alpha V_\alpha \otimes M_\alpha,$$

with $d_\alpha = \dim V_\alpha$ and $m_\alpha = \dim M_\alpha$, then

$$\mathcal{A}_{r,b}^{\mathrm{phys}} \cong \bigoplus_\alpha I_{V_\alpha} \otimes B(M_\alpha), \qquad p_{r,\alpha} = \frac{d_\alpha m_\alpha}{\sum_\beta d_\beta m_\beta}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

OPH quotient ensemble. An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \to \mathbb{R}_{>0}, \qquad S_r : Q_r \to \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q)e^{-S_r(q)}, \qquad Z_r = \sum_{q \in Q_r} w_r(q), \qquad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior ν_r on Q_r and set $\mu_r = (n_r)_\# \nu_r$. Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

Normal-form projector non-selection. For any map $N : Q \to Q_{\mathrm{nf}}$, the induced map on laws

$$\mathcal{C}_Q(\mu) = N_\# \mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on Q_{nf} is fixed. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

Selection-gap corollary. Let $X \subseteq Q_{\mathrm{nf}}$ be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine X and its quotient-visible invariants, but they do not choose a member of X. If two laws μ, ν are supported on X and concentrate on different candidates, both are fixed by $\mathcal{C}_Q$. Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

Finite MaxEnt quotient ensemble. For finite Q, positive m, and quotient observables $F_1, \ldots, F_k$, maximizing

$$\mathcal{H}_m(\nu) = -\sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \qquad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state σ_r,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\operatorname{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

The finite constraint ledger must name every $F_{r,a}$, its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

Refinement compatibility and RG closure. For $s \succeq r$, let $c_{sr} : Q_s \to Q_r$ be the physical coarse map. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q' : c_{sr}(q') = q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant $\alpha_{sr} > 0$ independent of q. Then

$$(c_{sr})_{\#} \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#} \mu_{k+1} - \mu_k\|_{\mathrm{TV}},$$

then

$$\|(c_{nr})_{\#} \mu_n - \mu_r\|_{\mathrm{TV}} \leq \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} \left[\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r\right]$$

to equal $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$. If the residual is uniformly bounded by ε, the induced total-variation defect is bounded by $\tanh \varepsilon$.

Implementation invariance and representative lifting. If implementations A, B have quotient bijections $h_r : Q_r^A \to Q_r^B$ satisfying

$$m_r^B(h_r q) = m_r^A(q), \qquad S_r^B(h_r q) = S_r^A(q), \qquad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_\# \mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas $A \mapsto A \otimes I_{\mathrm{anc}}$. It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\widetilde{\mu}_r(x) = \mu_r(\pi_r x)\kappa_r(x \mid \pi_r x), \qquad \sum_{x:\pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then $(\pi_r)_\# \widetilde{\mu}_r = \mu_r$. Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

Quotient-lumpable kernels and sampler correctness. A representative kernel $\widetilde{P}(x, y)$ descends to Q_r only when

$$P_Q(q, q') = \sum_{y:\pi(y)=q'} \widetilde{P}(x, y)$$

is independent of the chosen representative $x \in \pi^{-1}(q)$. For $w(q) = m(q)e^{-S(q)}$ and proposal $R(q, q')$ with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min\left\{1, \frac{w(q')R(q', q)}{w(q)R(q, q')}\right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

Repair generators are not selectors. A repair generator of the form

$$L_{\mathrm{rep}} = \sum_C c_C(I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations E_C are defined on $L^2(X_r, \pi_r)$, so the reference law π_r is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a positive lower bound γ_*, then

$$L_{\mathrm{rep}} \geq \gamma_* \kappa_r (I - P_0).$$

Finite repair completeness gives $\kappa_r > 0$ at fixed regulator. A uniform refinement lower bound $\inf_r \kappa_r > 0$ is a separate theorem or receipt.

Finite evidence accuracy. For bounded coarse observables O, if

$$\|\widehat{\mu}_s - \mu_s\|_{\mathrm{TV}} \le \epsilon_{\mathrm{samp}}$$

and the refinement defects sum to ϵ_{ref}, then

$$\left|\mathbb{E}_{\widehat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O]\right| \le 2\|O\|_\infty(\epsilon_{\mathrm{samp}} + \epsilon_{\mathrm{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

Vacuum promotion gate. A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. Vacuum promotion requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance $J_r(q, q') = J_r(q', q) \ge 0$, local potential V_r, and slab thickness $a_{t,r}$ are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q')(f(q) - f(q')) + V_r(q)f(q)$$

is self-adjoint and bounded below on $L^2(Q_r, m_r^0)$; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state Ω_r, and the finite vacuum law is

$$\mu_r^{\mathrm{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$, the Doob kernel is stochastic and detailed-balanced with μ_r^{vac}. Continuum promotion additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

Primordial and cosmological prediction firewall. A screen covariance is not automatically a primordial curvature spectrum. The map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has a null space unless a radial prior, finite parametric family, or source-stress bridge is declared. OPH primordial promotion requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, screen-to-radial-lift, radial-null-space, and forward-projection receipts. Observable CMB promotion further requires frozen source, solver, likelihood, no-data-use, pooled-reducer, and falsification-rule receipts.

Claim tiers and required receipts. Every ensemble-facing run records immutable ensemble id, claim tier, regulator id, representative schema hash, gauge action hash, canonicalizer hash, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler hash, smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt, not to the ensemble definition. The claim tiers are

$$\begin{aligned} E0 &: \text{seed noise, proposal noise, repair jitter,} \\ E1 &: \text{conventional reference ensemble,} \\ E2 &: \text{OPH-native quotient ensemble,} \\ E3 &: \text{OPH vacuum,} \\ E4 &: \text{OPH primordial field,} \\ E5 &: \text{observable cosmological prediction.} \end{aligned}$$

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of S_r unless explicitly declared.

3 E5 Prediction Contract

An observable cosmological prediction requires all source artifacts to be fixed before likelihood comparison. The minimum contract is:

1. immutable hashes for OPH source artifacts, schema versions, and regulator manifests;
2. immutable scale-bridge hashes for the generation, geometry, background, clock, scale certificate, source embedding, mode basis, mode lineage, boundary condition, solver convention, freezeout surface, and source DAG;
3. immutable hashes for Boltzmann solver source, version, tolerances, inputs, and build environment;
4. immutable hashes for likelihood code, datasets, masks, covariances, and nuisance-prior definitions;
5. a no-data-use manifest showing that likelihood values did not enter source artifact generation;
6. global pooled reducers for nonlinear source estimates;
7. a fixed falsification rule and reporting template before executing the likelihood.

A source-only anomaly abundance belongs to the source-artifact layer. It must be frozen before the Boltzmann run and before likelihood comparison.

Definition 3.1 (proof-carrying data artifact)**.** *A data-facing artifact is not represented by a value plus a receipt Boolean. It is a typed object*

$$\mathfrak{A}_v = (id_v, \tau_v, x_v, \mathcal{X}_v, \mathcal{U}_v, Par(v), \mathcal{D}_v, \mathcal{E}_v, h_v^{\mathrm{thm}}, h_v^{\mathrm{wit}}, h_v^{\mathrm{build}}, h_v^{\mathrm{art}}),$$

where τ_v is the semantic type, $\mathcal{X}_v$ records axes, domain, grid, and support, $\mathcal{U}_v$ records units, gauge, frame, pivots, and conventions, $Par(v)$ is the complete parent set, $\mathcal{D}_v$ is the data-class ledger, $\mathcal{E}_v$ is the error envelope, and the four hashes commit to the theorem version, witness, build environment, and artifact payload. All hashes are content-addressed commitments, not display labels.

Definition 3.2 (scale-bridge cross-receipt manifest)**.** *The physical scale bridge supplies a nonempty cross-receipt manifest. Every child receipt repeats the same regulator-family id, generation id, geometry hash, background hash, clock hash, scale-certificate hash, source-embedding hash, mode-basis hash, mode-lineage hash, boundary-condition hash, solver-convention hash, freezeout-surface hash, and source-DAG hash. An empty manifest or a missing identity field fails before likelihood execution.*

Theorem 3.3 (no receipt-Boolean promotion)**.** *Let G be the finite acyclic graph of data-facing artifacts. If every node verifier recomputes its predicate from its artifact, witness, complete parent set, data ledger, and error envelope, then a passed* PhysicalCMBPrediction *node implies that every required source, solver, likelihood, and no-data-use parent predicate also holds. In particular, caller-supplied receipt Booleans cannot promote an artifact across a semantic type boundary.*

Proof. Topologically order G. A root passes only when its verifier recomputes the corresponding predicate from the committed artifact and witness. A non-root node additionally requires every parent to pass. Induction gives all parent predicates and the local predicate at each passing node. A Boolean supplied by the producer is merely input data unless it is itself recomputed by such a verifier. $\square$

The promotion order used by this paper is

$$\begin{aligned}\textsc{DiagnosticProxy} \prec \textsc{FiniteScreenSource} \prec \textsc{FinitePrimordialSource} \\ \prec \textsc{PhysicalBoltzmannSource} \prec \textsc{PhysicalCMBPrediction}.\end{aligned}$$

No simulator export may jump over an intermediate node by relabelling a diagnostic scalar as a physical function.

The coherent-matter same-channel theorem fixes a local scalar-channel identity:

$$S^{\mathrm{can}}_{\mathrm{coh}} = \mathbf{1}_{\text{self-read}} R_U P_U C_U \in \mathcal{E}_{r,C}$$

for nondegenerate OPH-coherent material sources under Scalar Edge-Center Exhaustion. That identity is insufficient for a physical dark/anomaly row in a likelihood contract. Physical promotion also requires the finite covariant parent to emit w_A, $c^2_{s,A}$, σ_A, Q^μ_A, exchange-current closure, recipient stress, gauge-independent variables, CDM-limit recovery, and refinement convergence.

4 Finite Primordial Source Artifact

The CMB-facing source object is

$$\mathfrak{S}^{\mathrm{prim}}_r = (\mathcal{Q}_r, n_r, \mu_r, q_r, J_r, \Pi_r, K_r, M_r, \Xi_r, \Sigma_r, E^{\mathrm{rel}}_r, A_{q,r}, \theta_r, q_{\mathrm{IR},r}, t_{\mathrm{IR},r}, \Sigma_{\star,r}, W_r, \Delta^2_{\zeta,r}, \mathcal{P}_{IJ,r}, \mathcal{E}_r).$$

Here $\mathcal{Q}_r$ is the quotient state space, n_r the normal-form map, μ_r the source law, q_r the quotient-visible scalar source, J_r the weighted carrier-to-harmonic readout, Π_r the projector removing gauge, monopole, and dipole directions, K_r a positive precision operator, M_r a mobility or repair-response operator, Ξ_r the stochastic forcing covariance, and $\Sigma_r = \mathrm{Cov}_{\mu_r}(q_r)$. The remaining entries record release energy, amplitude, spectral exponent, infrared suppression, freezeout cut, radial window, lifted primordial spectrum, species covariance, and errors.

The old shorthand "screen count plus fitted scalar curve" is therefore only a diagnostic proxy. It does not determine amplitude, tilt, infrared suppression, species covariance, or physical Boltzmann inputs without the source artifact above.

5 Comparable Data Lanes

Lane	Allowed diagnostic output	E5 promotion blocker
CMB TT/TE/EE/lensing	Binned overlays, CAMB/CLASS scaffolds, peak metrics	Missing frozen source, Boltzmann, official likelihood, or no-data-use receipt.
BAO/SN/H_0	Distance-ladder and background-expansion diagnostics	Missing physical scale bridge and frozen background source functions.
Weak lensing and growth	Diagnostic S_8, growth, and lensing comparisons	Missing global stress closure, dark/anomaly kernels, and covariance-ready likelihood.
Large-scale structure	Prototype transfer/growth tables	Missing source-only primordial and growth-seed receipts.
Visualizer payloads	Observer cameras, screen fields, object/worldline diagnostics	Missing claim receipts; visual rendering is not a likelihood.

6 Simulator Contract

Every data-facing export should carry:

- `NO_DATA_USE_RECEIPT`
- `SOURCE_PROVENANCE_RECEIPT`
- `POOLED_SOURCE_REDUCER_RECEIPT`
- `BOLTZMANN_TRANSFER_RECEIPT`
- `FROZEN_LIKELIHOOD_RECEIPT`
- `PHYSICAL_CMB_PREDICTION_RECEIPT`

The receipt may be false. False is preferable to silently promoting a diagnostic curve.

7 Adversarial Contract Rejections

A conforming simulator contract must fail closed on the following cases.

1. negative or out-of-domain source parameters such as $q_{\mathrm{IR}} < 0$, missing freezeout surfaces, one-character hashes, and 1×1 zero kernels;
2. child artifacts declaring `source_only=true` while any transitive parent is measurement-fit, likelihood-fit, hand-tuned after looking at data, or outside the hermetic read set;
3. local shard averages used as nonlinear global reducers;

4. defaulting a single source row to a global pooled reducer when no pooled reducer receipt was constructed;
5. accepting arbitrary labels such as "anomaly" or "recipient" as physical stress rows without a finite covariant parent and stress/exchange closure receipt;
6. promoting temporal repair gaps, spatial refinement exponents, primordial tilt, and physical exchange rates as if they were the same semantic type.
7. accepting a source-only ρ_A or Ω_A claim unless `RHO_A_SOURCE_RECEIPT` passes;
8. accepting a `RHO_A_SOURCE_RECEIPT` whose receipt decomposition is anything other than `RHO_A_TRANSPORT_RECEIPT` plus `ANOMALY_ABUNDANCE_SOURCE_RECEIPT`;
9. accepting an `ANOMALY_ABUNDANCE_SOURCE_RECEIPT` whose load observable is computed from an amplitude-loaded parent occupation field normalized by an external Ω_A, Planck residual, SPARC fit, cluster fit, Boltzmann residual, or likelihood comparison;
10. accepting an anomaly abundance selector whose source DAG reads CMB, BAO, supernova, weak-lensing, RSD, SPARC, cluster, posterior, likelihood, nuisance-fit, or diagnostic residual data;
11. accepting a selector whose release law lacks a quotient-intrinsic ensemble law or source MaxEnt constraint ledger;
12. accepting a selector whose refinement defect exceeds the declared `LOAD_REFINEMENT_COMPATIBILITY` tolerance;
13. accepting a selector whose hidden carrier coordinates, port labels, repair schedules, or worker metadata change L_A.

The no-data-use verifier must traverse the complete parent graph, including file reads, network reads, environment variables, configuration values, caches, random seeds, dependency versions, and human-selected branch identifiers. A missing channel is a blocker.

8 Unclosed Claim Boundaries

1. Implement the finite primordial source artifact schema and verifier in the simulator.
2. Specify official likelihood readiness for Planck, ACT, BAO, SN, weak lensing, and growth lanes.
3. Make pooled reducers the only allowed source of nonlinear global quantities.
4. Define the falsification thresholds before running any public comparison.